



Improving Weather Predictions in Charlottesville

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Agenda

1. Motivation
2. Assumptions
3. Methodology
4. Modeling
5. Results/Future improvements

Weather in Charlottesville is an unpredictable stochastic system



Mar. 11, 2026

Temperatures soared
over 80 F



Mar. 12, 2026

Snow and hail occurred
during the afternoon



Mar. 16, 2026

Rain with a tornado
threat

**Why is
prediction
accuracy
important for
weather?**

- Reduces public confusion about weather conditions and forecasts
- Decreased forecasting accuracy erodes public trust
- Decreased last-minute operational changes

Weather forecasting frameworks can be improved and more accessible via increased state granularity in DTMC's

Current Approaches:

Physics-based systems with **ODEs** and **PDEs** to simulate atmospheric changes

Treating the atmosphere like a **fluid** and modeling its **movement** to predict **large-scale** weather events

Complex **machine learning** algorithms for data-driven weather prediction

Our Approach:

Create a **DTMC**-based stochastic model to capture **granular** states

Use a **fine-grained** state representation to better capture **extreme** and rapidly changing **conditions**

Make an **interpretable** framework for analyzing weather **dynamics** and **transition behavior**

25 Years of Charlottesville Weather Data was Used to Develop a Weather Prediction DTMC

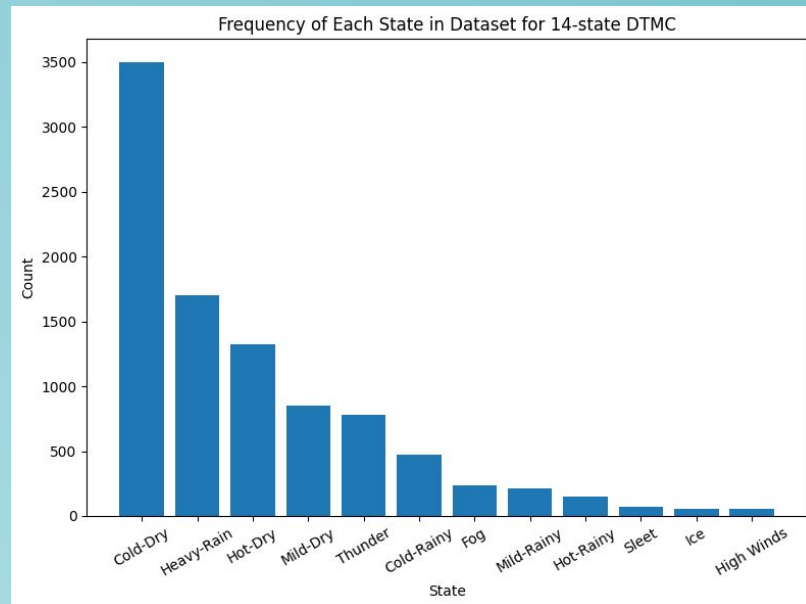
Daily Features:

- Min / Max Temperature
- Min / Max Precipitation
- Weather Event Flags:
 - Snow, Fog, Ice, Hail, Sleet, and Thunder

Classification of States:

1. Weather Event Present
2. High Rainfall (Above 75th Percentile)
3. Otherwise based on:
 - a. Precipitation & Temperature Bins

25-state Configuration: Combines discretized temperature, precipitation, and grouped severe weather events at higher granularity



Our Weather Prediction DTMC was Fitted Based on Several Assumptions



Location

Based on Charlottesville data; methodology generalizes to other locations



Discretization

Weather can be separated into discrete states



First-Order

Weather tomorrow will only depend on today's weather



External Factors

Effect of atmospheric conditions and other external factors are negligible



Stationary

Training and testing follow the same distribution



Data

Weather measurements are consistent and comparable over time

3 Models Were Tested and Analyzed to Improve Prediction Accuracy Metrics

Data:

Training Data: 1/1/2000-12/31/2024

- Used to calculate transition probabilities

Testing Data: 1/1/2025-12/31/2025

Models Tested:

1. Full DTMC (14 states, 25 states)
2. Seasonal DTMC (14 states, 25 states)
3. Second-Order DTMC (14 states, 25 states)

Evaluation Metrics

General Prediction Accuracy:

- Does the predicted next state match the actual next state?

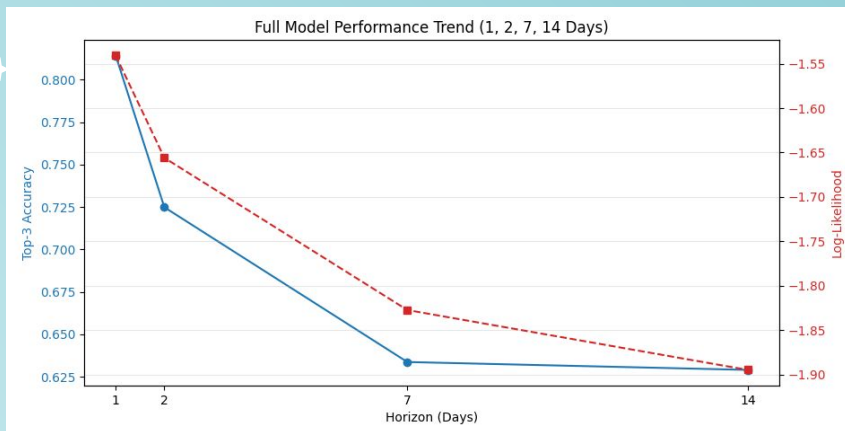
Top-3 Accuracy:

- Is the actual next state in the top 3 predicted states?

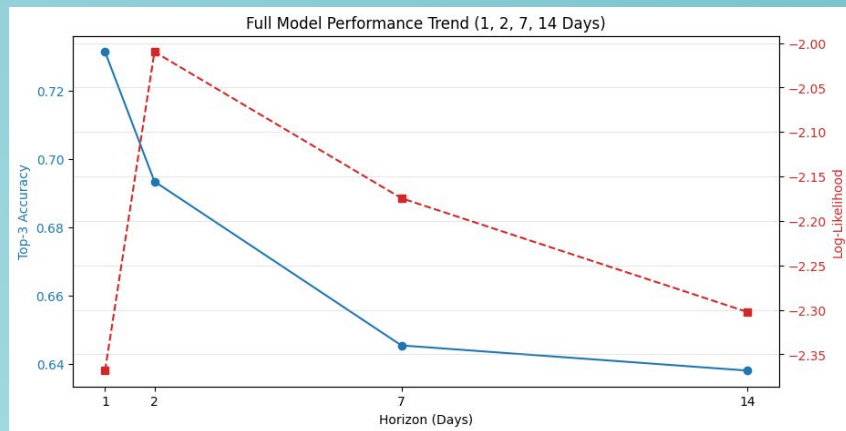
Log-Likelihood:

- How confidently do transition probabilities represent real weather dynamics?

14-state Full DTMC Achieves Higher than 80% Top-3 Accuracy for Next-Day Weather



14-State DTMC



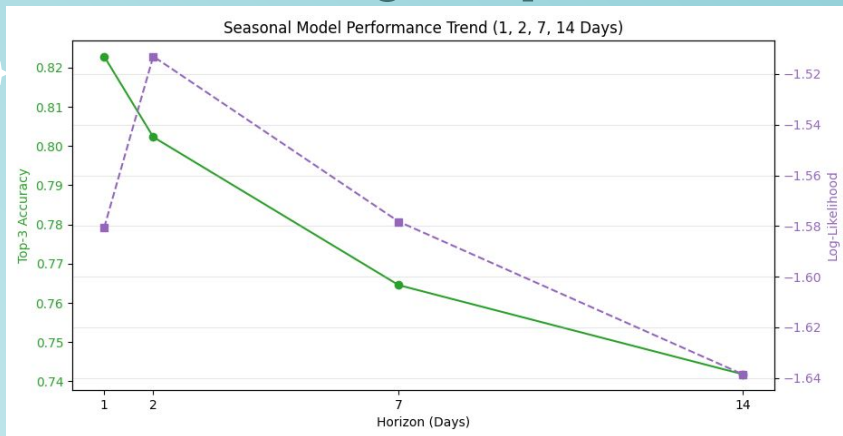
25-State DTMC

Accuracy decreases overtime as horizon window increases

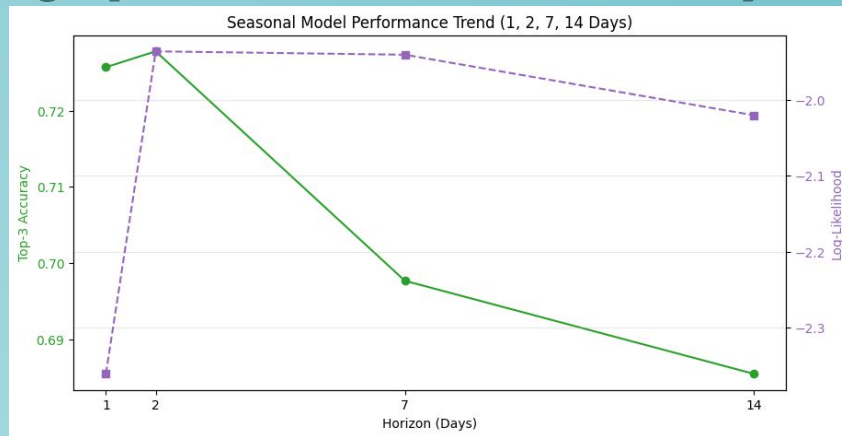
Log-likelihood for the 25-state DTMC is always less than the 14-state DTMC

Top-3 Accuracy for 25-state DTMC is better than the 14-state DTMC after 7/14 days

Seasonal 14-State DTMC has More Than 82% Next-Day Top-3 Accuracy (+2% vs. Full DTMC)



14-State DTMC

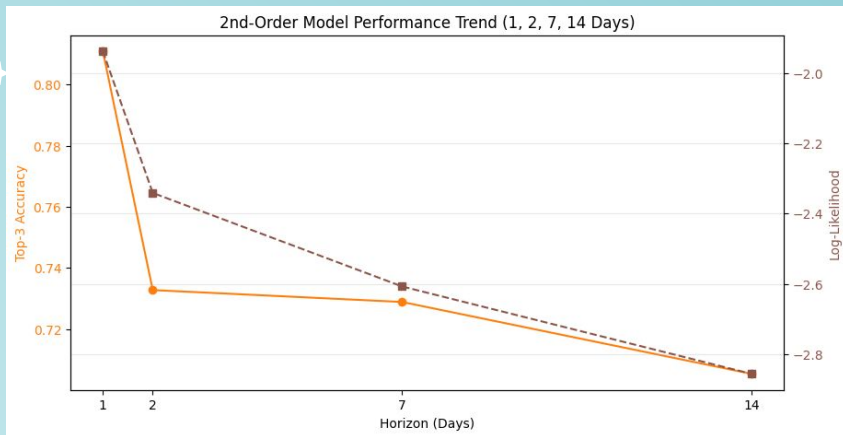


25-State DTMC

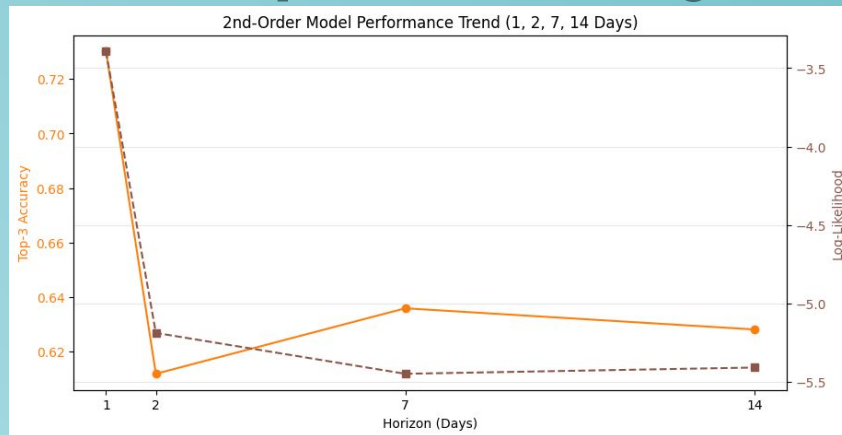
Seasonal DTMC in both states have higher Top-3 Accuracies than their full DTMC counterpart

Top-3 Accuracy and Log-likelihood for the 25-state DTMC is always less than the 14-state DTMC

2nd-Order DTMC: No Gain Over Seasonal, But Shows Gains First-Order in Top-3 Accuracy



14-State DTMC



25-State DTMC

2nd-order DTMC shows a sharp drop in day-2 accuracy, but a smoother decline in top-3 accuracy over longer horizons

2nd-order DTMC has lower log-likelihood than seasonal and 1st-order models

25-state DTMC continues to underperform relative to the 14-state model

Results / Future Improvements

Takeaways:

- The 14-state Seasonal DTMC performed best overall
 - Top-3 Accuracy: 82%-74%
 - Log-likelihood: -1.52 to -1.64
- Model top-3 accuracy always decreased as the horizon window increased
- Increasing state complexity doesn't significantly improve Top-3 Accuracy or Log-likelihood

Future Work:

- Reduce training dataset to 2010-2024 to see if climate change effects influence prediction accuracies
- Implement a 2nd Order DTMC for the Seasonal Model and higher-order DTMCs
- Expand dataset features to include atmospheric conditions and other external factors

Additional Sources

[1] IBM, "Weather Models," *Ibm.com*, Jun. 29, 2023.
<https://www.ibm.com/think/topics/weather-models>

[2] "The National Academies Press | Mathematics and Statistics of Weather Forecasting,"
nap.nationalacademies.org.
<https://nap.nationalacademies.org/resource/other/deps/illustrating-math/interactive/mathematics-and-statistics-of-weather-forecasting.html>

[3] "Search | Climate Data Online (CDO) | National Climatic Data Center (NCDC)," *www.ncei.noaa.gov*. <https://www.ncei.noaa.gov/cdo-web/search?datasetid=GHCND>